

Guillermo A. Fidalgo Rodríguez

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About Me

I'm a lifelong learner and I'm also passionate about software training, education and outreach. I am in the training team of [IRIS-HEP](#), [HSF](#), and a certified instructor of [The Carpentries](#). I have done several outreach activities with high school teachers and students in Puerto Rico. I've organized and mentored many workshops teaching at different academic levels from high school students to new HEP Postdocs on how to use basic software tools and do analysis. Most recently, I've mentored interns in the [U.S. CMS PURSUE](#) program to increase participation from underrepresented groups in HEP. Additionally, I've also facilitated the 2023 and 2024 CMS Data Analysis School at the LHC Physics Center at Fermilab (CMSDAS@LPC). My current interest is Machine Learning (ML) and its applications. I've worked on using ML to monitor the data quality of CMS Tracker and its future upgrade. I've participated and won a Machine Learning Hackathon organized by Lockheed Martin using Transfer Learning to make use of a previously [trained CNN](#) with the task to classify surface images from Mars Curiosity rover.

Education

PHD | UNIVERSITY OF ALABAMA | 2024 – PRESENT

- Developing analysis framework for the “four top quark production to single lepton final state with Run3” analysis
- Setup the Multi Module Testing System (MMTS) for the testing HGCAL hexabboards
- Contribute to HGCAL DQM software development

MASTER'S | UPRM | 2020-2024

- Trigger Studies for Emerging Jets Analysis and Machine Learning for Tracker DQM at CMS ([Master Thesis](#))
- Contribute to ML for Tracker DQM efforts

UNDERGRADUATE | UNIVERSITY OF PUERTO RICO MAYAGUEZ (UPRM) | 2014-2020

- Major: Physics
- Minor: Applied Mathematics
- President/Founder of the Scientific Software Club at UPRM

Experience

PUBLISHED PAPERS

- D. Julson, E. Reinhardt, A. Krutsylo, R. Sohal, G. Fidalgo, S. Gleyzer, and E. Usai, *Continual Learning via Ensemble-Based Depth-Wise Masked Autoencoders for Data Quality Monitoring in High-Energy Physics* - [arXiv:2603.02369](#) [[hep-ex](#)]
- “Search for Emerging Jets with full Run 2 data” - [https://doi.org/10.1007/JHEP07\(2024\)142](https://doi.org/10.1007/JHEP07(2024)142)
- “Broadening the scope of Education, Career and Open Science in HEP” - [arXiv:2203.08809](#) [[physics.ed-ph](#)]
- “Particle Physics Outreach to K-12 Schools and Opportunities in Undergraduate Education” - [arXiv:2203.10953](#) [[physics.ed-ph](#)]
- “Facilitating Non-HEP Career Transition” - [arXiv:2203.11665](#) [[physics.ed-ph](#)]
- “U.S. CMS - PURSUE (Program for Undergraduate Research SUMmer Experience)” - [arXiv:2209.10109](#) [[physics.ed-ph](#)]
 - Acknowledged for tutoring interns about HEP software tools, for effort in introducing HEP to a wider audience and take a small step towards trying to enhance the diversity of the HEP community.

SOFTWARE TRAINING AND OUTREACH ACTIVITIES

- **Mentored in the following:**
 - HSF/IRIS-HEP training @ Brookhaven National Laboratory (In person) **June 3-5, 2026** [[Link](#)]
 - HSF/IRIS-HEP Software Basics Training (Virtual) **September 28-30, 2022** [[Link](#)]
 - “DANCE@Snowmass” (University of Washington, Seattle, USA) **July 19-23, 2022** [[Link](#)]
 - “Software Carpentry” (Virtual) **July 13-15, 2022** [[Link](#)]
 - “Matplotlib Training” (Virtual) **April 21-22, 2022** [[Link](#)]
 - “Software Carpentry” (Virtual) **March 28-30, 2022** [[Link](#)]
 - “Data Analysis for Lab Research” (UPRM, Puerto Rico) **March 5, 2022** [[Link](#)]
 - “Software Carpentry” (Virtual) **Dec 13-15, 2021** [[Link](#)]
- **Developed, organized and mentored the following:**
 - “Python 101 for STEM Teacher” (UPRM, Puerto Rico) **August 20, 2022** [[Link](#)]
 - “Python 101 for STEM Teachers @CROEM” (CROEM, Mayagüez, Puerto Rico) **August 17, 2022** [[Link](#)]
 - “U.S. CMS – PURSUE” (Virtual) **June 6 – Aug 22, 2022** [[Link](#)]
 - Mentored undergraduate students on HEP Software
 - “Astronomy Course for CROEM high school students” (Virtual) **Jan - May 2021** Complemented the programming side of an introductory course of Physics and Astronomy with Python programming and exercises for local high school students.
 - “2020 STEM Teachers Workshop” (Virtual) **15-16 July 2021** Workshop dedicated to enabling K-12 teachers with python coding language. [[Link](#)]

- "Machine Learning Basics for STEM Teachers" (Virtual) **February 13-14, 2021** [\[Link\]](#)
- "Data Analysis for STEM Teachers" (Virtual) **July 15-16, 2020** [\[Link\]](#)
- "Introduction to python programming for Undergrads" (UPRM, Puerto Rico) **22 February 2020** Workshop intended to teach local Physics students with python coding language with HEP applications. [\[Link\]](#)
- "Scientific Software Club (SSC)" **Founded in August 2019** Formed the [Scientific Software Club](#) at UPRM to bring together students interests in software related research projects. Have given around 5 workshops including topics: Bash, Git and Python. Currently have over 100 members and growing.
- "STEM Teachers Workshop" (UPRM, Puerto Rico) **3-4 June 2019** Workshop dedicated to enabling high school STEM teachers with coding. [\[Link\]](#)

CONFERENCE PRESENTATIONS

1. *"Four top quark production at Run3 in the single-lepton final stat with CMS"* at APS Global Physics Summit (Denver, Colorado) **March 19, 2026** [\[Link\]](#)
2. *"Teaching Python the Sustainable Way: Lessons Learned at HSF Training"* at PyHEP 2022 Workshop (Virtual) **September 12, 2022** [\[Link\]](#)
3. *"PyROOT tutorial experience from SWC Workshop"* at 12th ROOT Users' Workshop (Virtual) **September 12, 2022** [\[Link\]](#)
4. *"New Trigger Studies for Emerging Jets at CMS Experiment"* at PRIMs/JTM Conference (Humacao, Puerto Rico) **April 9, 2022** [\[Link\]](#)
5. *"Recent Progress in ML for Tracker DQM"* at DPF Conference (Virtual) **July 12, 2021** [\[Link\]](#)
6. *"Machine Learning in DQM at CMS Experiment"* at Physcon Conference (Rhode Island, USA) **November 13-17, 2019** [\[Link\]](#)
7. *"Machine Learning and Deep Neural Networks at CMS Experiment"* at Physics symposium (University of Puerto Rico Mayagüez) **May 24, 2019** [\[Link\]](#)
8. *"Machine Learning in DQM at CMS Experiment"* at PRIMs/JTM Conference (Mayagüez, Puerto Rico) **May 4, 2019** [\[Link\]](#)
9. *"Using ML techniques for DQM at CMS"* at ML Hackathon (University of Puerto Rico Mayagüez) **April 25, 2019** [\[Link\]](#)
10. *"Machine Learning in DQM at CMS Experiment"* at FNAL 51st Annual Users Meeting and New Perspectives Meeting (Fermilab, Batavia, IL, USA) **June 18, 2018** [\[Link\]](#)

Leadership

TREASURER | SOCIETY OF PHYSICS STUDENTS (UPRM-CHAPTER) | JANUARY 2017-2018

- Treasurer of the SPS UPRM Chapter - in charge of managing the chapter's finances, sales, and membership enrollment.

SPS DEMONSTRATIONS COMMITTEE

- I have experience presenting to local high school students and to the general public essential topics in physics such as: Conservation of Energy, Conservation of Momentum, Torque, Atmospheric Pressure and Electromagnetism.

PRESIDENT/FOUNDER | SCIENTIFIC SOFTWARE CLUB | AUGUST 2019 – 2020

- In the summer of 2019, I founded the first software society registered under the physics department. This club is focused on education and application of software used in STEM research in hopes that it will complement every physics undergrad's academic career.
- Trained students in Linux, Git and Python via workshops focusing on best practices with material inspired by [Software Carpentries](#)
- Website - [Scientific Software Club - UPRM](#)

Skills & Abilities

- Proficient in Python and data analysis packages like numpy, pandas, matplotlib and Scikit-HEP ecosystem
- Proficient in making documents and presentations with *LaTeX*
- Proficient with Basha and Git
- Knowledgeable in CI/CD with GitHub Actions and Gitlab CI
- Fluent in English and Spanish
- Knowledgeable in web development with following tools:
 - Jekyll
 - Personal Website: <https://guillermofidalgo.github.io/>
 - Google Sites
 - U.S. CMS PURSUE: <https://sites.google.com/upr.edu/uscms-pursue/>
 - SSC Website: <https://sites.google.com/upr.edu/ssc-uprm/home>